

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 6.2

Convolution of Discrete Time Signals

Aim:

To compute and visualize the convolution of two discrete-time signals. Convolution is fundamental operation in discrete-time signal processing that combines two signals to produce a third signal, which is a modified version of one of the signals.

Program:

```
% Test Case: Convolution of Discrete-Time Signals
```

```
% Define the discrete time signals
```

```
x_n = [1, 2, 3]; % Signal x[n]
```

```
h_n = [1, -1]; % Signal h[n]
```

```
% Compute the convolution
```

```
y_n = conv(x_n, h_n); % Convolution of x[n] and h[n]
```

```
% Define the time index for the output signal
```

```
n_y = 0:(length(x_n) + length(h_n) - 2); % Length of y[n]
```

```
% Plot the input signals and their convolution
```

```
clf; % Clear the current figure
```

```
% Plot x[n]
subplot(3,1,1);
bar(0:length(x_n)-1, x_n, 'b');
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');

% Plot h[n]
subplot(3,1,2);
bar(0:length(h_n)-1, h_n, 'r');
xlabel('n');
ylabel('h[n]');
title('Signal h[n]');

% Plot y[n] (convolution result)
subplot(3,1,3);
bar(n_y, y_n, 'g');
xlabel('n');
ylabel('y[n]');
title('Convolution of x[n] and h[n]');
```

Output:



Result:

1. Plot of $x[n]$:
 - A stem plot showing the values of $x[n]$ for $n = 0, 1, 2$.
2. Plot of $h[n]$:
 - A stem plot showing the values of $h[n]$ for $n = 0, 1$.
3. Plot of Convolution Result $y[n]$:
 - A stem plot showing the values of $y[n]$, which is the result of the convolution operation.

This test case will help you understand the convolution operation for discrete-time signals and visualize the result using Scilab.